COMPUTER ENGINEERING

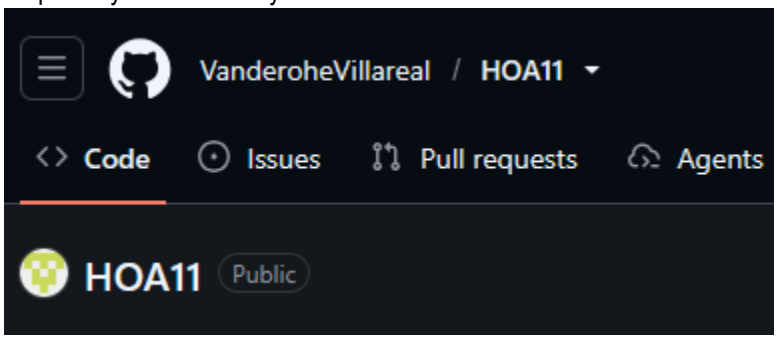
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COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/15/26	**DATE SUBMITTED:**	10/15/26
NAME:	Villareal, Vanderohe Vernon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 11.1 Containerization

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.



2. Install Docker and enable the docker socket.

Installation and Verification of active docker socket on each managed node.

Ubuntu

```
villareal1@server1:~$ sudo apt install -y docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
villareal1@server1:~$ sudo systemctl enable --now docker
villareal1@server1:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: e>
     Active: active (running) since Thu 2026-10-15 00:30:55 UTC; 58s ago
```

```
villareal1@server1:~$ sudo systemctl enable --now docker.socket
villareal1@server1:~$ sudo systemctl status docker.socket
● docker.socket - Docker Socket for the API
     Loaded: loaded (/usr/lib/systemd/system/docker.socket; enabled; preset: en>
     Active: active (running) since Thu 2026-10-15 00:30:53 UTC; 1min 18s ago
```

CentOS

```
[villareal@localhost ~]$ sudo systemctl enable --now docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /us
r/lib/systemd/system/docker.service.
[villareal@localhost ~]$ sudo systemctl enable --now docker.socket
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/li
b/systemd/system/docker.socket.
```

```
[villareal@localhost ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: d
     Active: active (running) since Thu 2026-10-15 08:55:06 PST; 27s ago
```

```
[villareal@localhost ~]$ sudo systemctl status docker.socket
● docker.socket - Docker Socket for the API
     Loaded: loaded (/usr/lib/systemd/system/docker.socket; enabled; preset: di
     Active: active (running) since Thu 2026-10-15 08:55:00 PST; 37s ago
```

3. Add a Docker group to your current user.

Adding a Docker group to the current user on each managed node.

Ubuntu

```
villareal1@server1:~$ sudo usermod -aG docker "$USER"
[sudo] password for villareal1:
```

CentOS

```
[villareal@localhost ~]$ sudo usermod -aG docker "$USER"
```

4. Create a Dockerfile to install web and DB servers.

```
villareal1@workstation: ~/HOA11
  GNU nano 7.2                          Dockerfile
FROM ubuntu:24.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y apache2 mariadb-server && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

COPY start.sh /start.sh

RUN chmod +x /start.sh

EXPOSE 80 3306

CMD ["/start.sh"]
```

villareal1@workstation: ~/HOA11

GNU nano 7.2 start.sh

```
#!/bin/bash
set -e

service mariadb start
exec apache2ctl -D FOREGROUND
```

5. Install and build the Dockerfile using Ansible.

deploy.yml

villareal1@workstation: ~/HOA11

GNU nano 7.2 deploy.yml

```
---
- name: Build and run the web and database container
  hosts: managed
  become: true
  tasks:
    - name: Create application directory
      ansible.builtin.file:
        path: /opt/activity11
        state: directory
        mode: "0755"

    - name: Copy Dockerfile
      ansible.builtin.copy:
        src: Dockerfile
        dest: /opt/activity11/Dockerfile
        mode: "0644"

    - name: Copy startup script
      ansible.builtin.copy:
        src: start.sh
        dest: /opt/activity11/start.sh
        mode: "0755"

    - name: Build Docker image
      ansible.builtin.command:
        cmd: docker build -t activity11-webdb:latest /opt/activity11

    - name: Remove previous container if it exists
```

```
    ansible.builtin.command:
      cmd: docker rm -f activity11-container
    failed_when: false
    changed_when: false

  - name: Run web and database container
    ansible.builtin.command:
      cmd: docker run -d --name activity11-container -p 8080:80 -p 3307:3306 >

  - name: Verify running container
    ansible.builtin.command:
      cmd: docker ps
    register: docker_status
    changed_when: false

  - name: Display container status
    ansible.builtin.debug:
      var: docker_status.stdout_lines
```

install_docker.yml

```
villareal1@workstation: ~/HOA11
 GNU nano 7.2                    install_docker.yml
---
- name: Install Docker on Ubuntu and CentOS
  hosts: managed
  become: true
  tasks:
    - name: Install Docker on Ubuntu
      ansible.builtin.apt:
        name:
          - docker.io
          - python3
        state: present
        update_cache: true
      when: ansible_os_family == "Debian"

    - name: Install DNF plugins on CentOS
      ansible.builtin.dnf:
        name: dnf-plugins-core
        state: present
      when: ansible_distribution == "CentOS"

    - name: Add Docker repository on CentOS
      ansible.builtin.command:
        cmd: dnf config-manager --add-repo https://download.docker.com/linux/ce>
        creates: /etc/yum.repos.d/docker-ce.repo
      when: ansible_distribution == "CentOS"

    - name: Install Docker on CentOS
      ansible.builtin.dnf:
```

```
        name:
          - docker-ce
          - docker-ce-cli
          - containerd.io
          - docker-buildx-plugin
          - python3
        state: present
      when: ansible_distribution == "CentOS"

    - name: Enable and start Docker
      ansible.builtin.service:
        name: docker
        enabled: true
        state: started

    - name: Enable Docker socket
      ansible.builtin.service:
        name: docker.socket
        enabled: true
        state: started
```

site.yml

```
villareal1@workstation: ~/HOA11
  GNU nano 7.2                                site.yml
---
- import_playbook: install_docker.yml
- import_playbook: deploy.yml
```

Running of the playbook (NOTE: due to different root passwords the Ubuntu and CentOS managed nodes have separate results)

```
PLAY RECAP *********************************************************************
centos-node                : ok=15   changed=5    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0
ubuntu-node                : ok=0    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
```

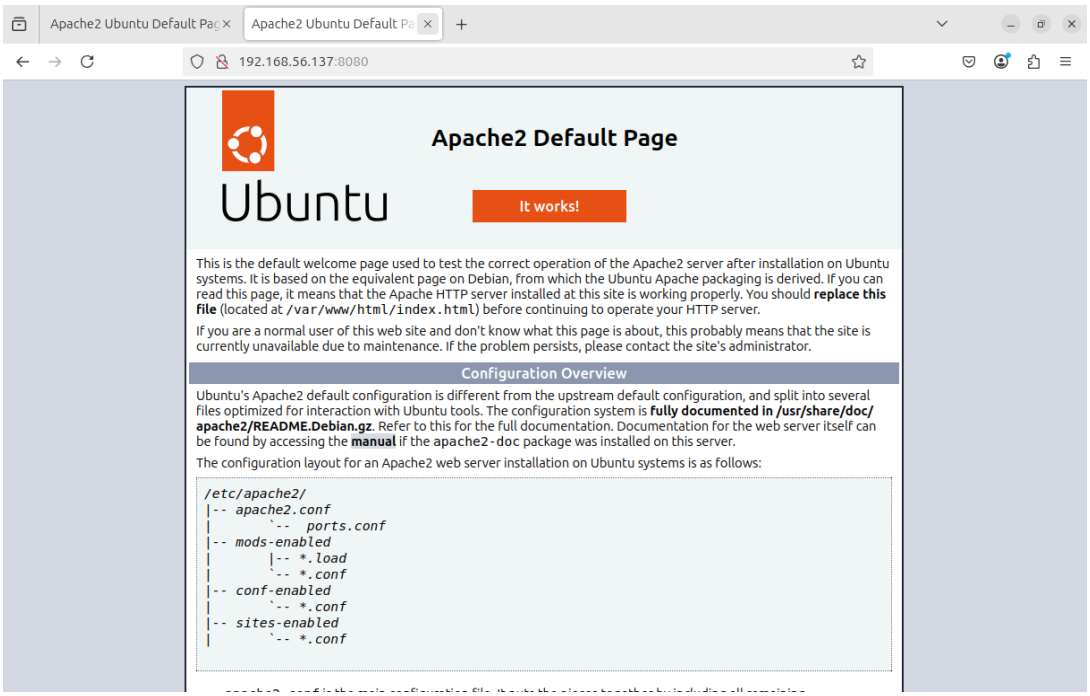
```
PLAY RECAP *********************************************************************
centos-node                : ok=0    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
ubuntu-node                : ok=13   changed=5    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
```

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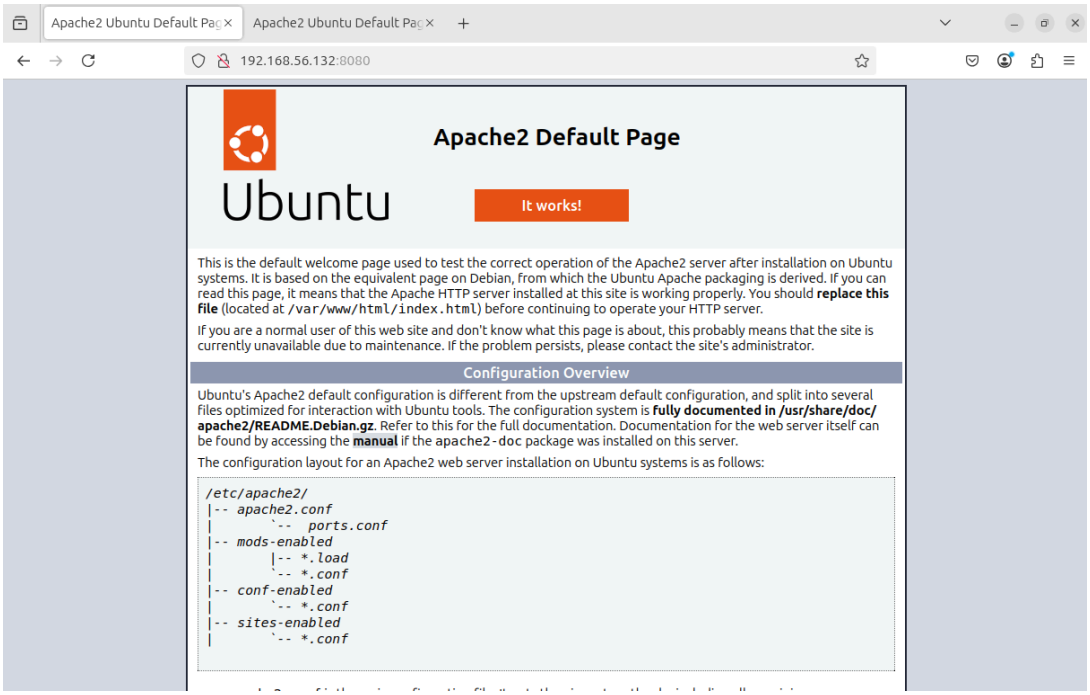
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Website Verification

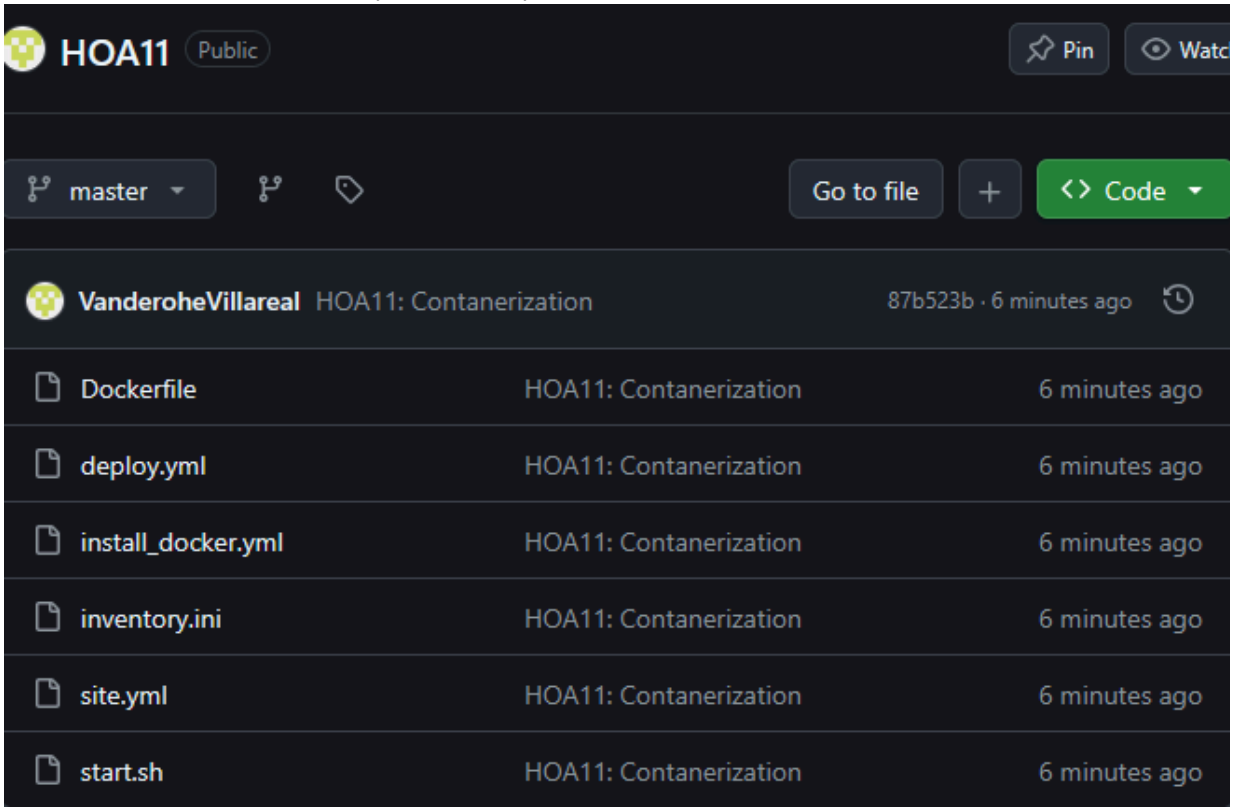
Ubuntu



CentOS



6. Add, commit and push it to your repository.



2. REFLECTIONS AND CONCLUSIONS

Reflections:

Answer the following:

1. What are the benefits of implementing containerizations?
   - Containerization makes applications easier to install, run, and manage. It allows applications to run in isolated environments, helps maintain consistency across different systems, and makes deployment faster. It also improves resource efficiency because containers share the host operating system.

Conclusion:

Overall, this activity helped me understand how containerization and automation work together in application deployment. By using Docker, I learned how to create a Dockerfile and build containers that run a web server and database server. Using Ansible, I learned how to automate the installation of Docker and deploy containers on Ubuntu and CentOS managed nodes from a single control node. This activity also improved my understanding of managing different operating systems, configuring services, and troubleshooting deployment errors. Through this experience, I gained practical knowledge of how Docker and Ansible can make application deployment more organized, consistent, and efficient.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction, idea generation, explanation of new commands**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI is used generally for grammar correction on the reflection and conclusion, idea generation for the reflection, and explaining of new commands from the activity that is yet to be learned therefore I asked the AI to guide me which commands are applicable.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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